Regularization, Initial-State Representation, and Optimization Budget in a Controlled Audit of Physics-Informed Heat Inversion

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Abstract

Purpose: We examine how classical regularization, initial-state representation and neural optimization budget affect comparisons in joint recovery of constant diffusivity and an unknown initial temperature field.

Methods: A prospectively versioned one-dimensional heat benchmark used two initial fields, 10 or 50 noisy observations, and 20 independent observation/noise designs, shared across field families to form 40 cases. Four spectral inverse procedures varied regularization and six- or twelve-mode representations. Two hard-boundary physics-informed neural network procedures used fixed architecture, three initializations per case, and 200 or 1000 L-BFGS iterations after Adam. We separately assessed diffusivity, initial-field and full-field errors, with paired comparisons against cross-validated twelve-mode inversion (CV12).

Results: All 400 scheduled jobs completed. For the richer initial field with 10 observations, the longer-budget neural procedure had lower diffusivity and initial-field errors than CV12 in nine of ten designs each. With 50 observations, CV12 had lower full-field error in eight of ten designs. Several contrasts remained uncertain under pointwise bootstrap intervals. The six-mode representation imposed a 21.82% initial-field error floor for the richer pattern.

Conclusion: Comparative conclusions varied with the specified procedure, observation regime and recovery target. The results support a limited pipeline audit rather than a universal ranking. Spectral truth, unequal tuning opportunities and ten designs per condition limit generalization.

Keywords: Physics-informed neural networks; inverse heat conduction; joint parameter and state estimation; regularization; spectral methods; computational benchmarking.

1 Introduction

Estimating a physical transport coefficient from temperature observations is more demanding when the starting temperature distribution is also unknown. For the heat equation, each initial sine component decays at a rate proportional to diffusivity and the square of its spatial frequency. Consequently, estimating diffusivity and reconstructing the initial field are coupled tasks, and errors in the recovered coefficient, initial state and space-time field need not move together. A comparison based on only one of these outcomes can obscure important differences between inverse procedures.

Neither joint heat inversion nor neural recovery of unobserved states is new. Zhao, Banda and Guo developed a classical procedure for simultaneous recovery of diffusivity, a spatial source and the initial state, using boundary control, boundary observations and a distributed temperature snapshot. Their observation and excitation conditions differ from the sparse interior observations examined here. [1] Oliveira and colleagues studied initial-temperature reconstruction with several physics-informed learning formulations for a heat problem with specified diffusivity. [2]

Comparative evidence also predates the present study. Grossmann and colleagues compared PINNs with finite elements across forward PDE problems and distinguished solution construction from subsequent evaluation. Their results motivate explicit numerical references and cost definitions, but do not resolve the joint inverse problem studied here. [3] Zapf and colleagues compared PINN and finite-element estimation of apparent diffusivity from brain tracer images, demonstrating the relevance of training formulation in a noisy inverse application. Their treatment of initial and boundary information differs from our unknown-initial-field formulation. [4] Jekic and colleagues examined noisy inverse fluid problems using PINNs and a finite-element/optimization reference, with prescribed initial-condition data. Their work establishes a direct precedent for noisy inverse benchmarking, rather than evidence that such comparisons remain unexplored. Jekic et al., preprint v2 (2026)

These precedents motivate a more specific question: **for joint constant- diffusivity and unknown-initial-field heat inversion, how does the comparison between specified neural and classical procedures change when**

initial-state regularization, basis capacity and neural optimizer budget are made explicit? The purpose is to examine the sensitivity of comparative conclusions, rather than to introduce a new solver or establish a ranking of entire method classes.

We address this question with an analytically generated one-dimensional heat benchmark, two initial-state families and two observation counts. Four spectral inverse procedures distinguish absent, fixed and observation-selected regularization, and six- versus twelve-mode representations. Two hard-boundary PINN procedures retain different L-BFGS iteration budgets after the same Adam stage. Observations, realized noise and diffusivity bounds are shared across procedures within each case. Diffusivity, initial-state and full-field errors are evaluated separately, with dataset-level neural initialization summaries, paired contrasts and a complete outcome archive.

The simulation, procedures and primary analysis were versioned before generating the evaluation data. This prospective design reduces opportunities for selecting favorable cases after inspection; it is not formal preregistration. The spectral truth favors the classical representation, while classical cross-validation and fixed neural settings provide different tuning opportunities. The study therefore offers a controlled audit of specific pipelines, with explicit representation, tuning, uncertainty and scope limits. It does not claim the first joint heat inverse study or a universally preferable method.

2 Methods

2.1 Problem and simulation

We considered joint recovery of a constant diffusivity and an unknown initial temperature field in the one-dimensional heat equation,

$$u_t = \alpha u_{xx}, \qquad (x,t) \in (0,1) \times (0,1], \qquad u(0,t) = u(1,t) = 0, \qquad u(x,0) = f(x). \tag{1}$$

The domain and coefficients define a normalized synthetic benchmark; no material or experimental measurement units were assigned. Both the scalar diffusivity α and the initial field f were unknown to the estimators. Boundary values and the domain were known. The true diffusivity was 0.05. Two initial fields were used:

$$f_A(x) = \sin(\pi x) + 0.5\sin(3\pi x), \qquad f_B(x) = f_A(x) + 0.25\sin(7\pi x). \tag{2}$$

For mode amplitudes a_j^*, the analytical reference was

$$u^*(x,t) = \sum_j a_j^* \sin(j\pi x)\exp[-0.05(j\pi)^2 t]. \tag{3}$$

We sampled $N = 10$ or 50 paired space-time coordinates uniformly on the unit square. Observations were $y_i = u^*(x_i,t_i) + 0.01 z_i$, with independent standard normal z_i. The noise standard deviation was fixed in benchmark temperature units rather than scaled by each dataset's signal. There were ten independent observation/noise designs per observation count. Within each design, coordinates, realized noise and validation folds were reused for families A and B. Thus the study contained 40 dataset cases from 20 independent designs, with ten designs in each observation-count/family stratum. The two family cases within a design were not independent replications.

Data seeds were 6000–6009 for N=10 and 7000–7009 for N=50. No initial-field measurements, true mode amplitudes or clean observation values were supplied to the fitting or regularization-selection procedures. Truth was used to generate data and to score fitted outputs.

2.2 Spectral reference procedures

We represented the estimated temperature field by

$$\widehat{u}_M(x,t) = \sum_{j=1}^{M} a_j \sin(j\pi x)\exp[-\alpha(j\pi)^2 t]. \tag{4}$$

This representation satisfies the boundary conditions and heat equation exactly for its estimated diffusivity. For each candidate diffusivity we solved

$$\min_{\boldsymbol{a}} \ \frac{1}{N}\|B(\alpha)\boldsymbol{a} - \boldsymbol{y}\|_2^2 + \lambda \sum_{j=1}^{M} j^4 a_j^2, \qquad B_{ij}(\alpha) = \sin(j\pi x_i) e^{-\alpha(j\pi)^2 t_i}. \tag{5}$$

Conditional amplitudes were computed by augmented least squares, avoiding normal-equation formation. Diffusivity was constrained to [1e-5,1]. We evaluated the profiled objective at 1601 points equally spaced in log diffusivity, refined every detected interior grid-local minimum with bounded scalar minimization (log-coordinate tolerance 1e-11), and retained the lowest objective among the refined, grid and endpoint candidates. This search does not certify a global minimum. Its objective profile is not a confidence interval.

Four spectral procedures were evaluated: unregularized six-mode fitting (unreg6); six-mode fitting with fixed lambda=1e-6 (fixed6); and six- or twelve-mode fitting with observation-based regularization selection (CV6 and CV12). For CV6/CV12, candidates were {1e-10,1e-8,1e-6,1e-4,1e-2}. We used five balanced folds, generated with seed 30000 plus the data seed. Every candidate was fitted using only its training folds. Selection minimized pooled held-out squared error divided by the total observation count, with exact ties favoring smaller lambda. The selected candidate was refitted on all observations. Positive CV penalties ensure unique conditional amplitude estimates, including when the training set has fewer points than basis functions. They do not ensure joint identification of diffusivity and amplitudes.

The twelve-mode family can represent both simulated truths, while the six-mode family excludes the seventh component of family B. This is a deliberate representation control. The exact spectral representation and finite spectral truth favor the classical reference; the comparison does not make its prior information equivalent to that of the neural field.

2.3 Physics-informed neural procedures

We fitted a fully connected neural field

$$\widehat{u}_\theta(x,t) = x(1-x)g_\theta(x,t), \tag{6}$$

where g_θ had three hidden layers of 32 tanh units and a linear scalar output. Inputs were scaled to [-1,1]; linear layers used Xavier-normal weights with tanh gain and zero biases. The prefactor enforces zero boundary values exactly. No initial-condition loss was imposed. A learnable scalar r parameterized diffusivity by

$$\alpha(r) = \exp\{\log(10^{-5}) + [\log(1) - \log(10^{-5})]\,\mathrm{sigmoid}(r)\}, \tag{7}$$

initialized to alpha=0.2. We minimized

$$\mathcal{L} = \frac{10}{N}\sum_i (\widehat{u}_\theta(x_i,t_i) - y_i)^2 + \frac{1}{512}\sum_{k=1}^{512}(\partial_t \widehat{u}_\theta - \alpha\partial_{xx}\widehat{u}_\theta)^2(x_k^c, t_k^c). \tag{8}$$

Derivatives were computed by automatic differentiation. The 512 uniform collocation points, drawn with seed 314159, were fixed across all cases and initializations. An independent set of 2048 points, seed 271828, was retained for residual evaluation. This residual diagnostic was not a selection rule or a substitute for reference-based recovery errors.

Both neural procedures used 2000 full-batch Adam steps, with learning rates 0.001 for network weights and 0.01 for the diffusivity parameter. These were followed by one L-BFGS stage with either a 200-iteration cap (PINN-200) or a 1000-iteration cap (PINN-1000). L-BFGS used learning rate 1, strong-Wolfe line search, history size 50, gradient tolerance 1e-12, change tolerance 1e-14, and maximum evaluation settings 250 or 1250, respectively. A 600-second training limit was declared per fit. Initializations used seeds 101, 102 and 103 for every case and both budgets. Both budgets were retained without selecting one using recovery error. Network settings were fixed following development work; they were not subjected to a tuning search matching classical cross-validation.

2.4 Evaluation and statistical summaries

Diffusivity and initial-field errors were the co-primary reported outcomes. Full-field error was also mandatory. We calculated

$$E_\alpha = \frac{|\widehat{\alpha} - 0.05|}{0.05}, \qquad E_f = \frac{\|\widehat{u}(\cdot,0) - f\|_{L^2(0,1)}}{\|f\|_{L^2(0,1)}}, \qquad E_u = \frac{\|\widehat{u} - u^*\|_{L^2((0,1)^2)}}{\|u^*\|_{L^2((0,1)^2)}}. \tag{9}$$

Errors were evaluated by trapezoidal quadrature using 401 spatial and 401 temporal grid points and reported as percentages. A planned 801-point check was performed for every procedure/start on the first design in each stratum, without refitting or replacing the primary scores.

For each dataset and neural procedure, we first took the median error across the three initializations, separately for each outcome. This describes typical initialization performance. It is not an ensemble predictor, and its three

outcome medians need not correspond to one fitted network. Stratum summaries then describe the ten dataset scores. All individual initialization scores were retained.

The prespecified paired contrasts subtracted CV12 error from the corresponding neural dataset score within each design. Differences were expressed in percentage points; negative differences favor the neural procedure. We reported means, medians, maxima and strict numerical win/tie/loss counts. Pointwise 95% percentile bootstrap intervals for median differences used 10000 design-level resamples, with seed 42000+N. Families were analyzed separately. These are descriptive small-sample intervals, with no adjustment for the 24 contrasts and no unplanned significance tests. A median paired difference is not the difference between two marginal medians. Intervals spanning zero do not establish equivalence.

Completion was assessed before accuracy. Failures, timeouts, interruptions and bound hits were to be retained without replacement. Three-start summaries required complete neural triads, and intervals required all ten complete pairs in a stratum. The completed study met these requirements throughout.

2.5 Prospective freeze, computation and reproducibility

The implementation and protocol were committed as b9d2b3e. The freeze record was created on 4 November 2026 at 05:46:00 UTC and committed as dad8cd5 at 05:46:12 UTC, before collection began at 05:46:59 UTC and before the first data generation timestamp at 05:47:17 UTC. This was a prospective versioned protocol, not formal preregistration. Earlier pilots were development work and were not pooled into evaluation.

The run used CPU float64 computation on Windows, Python 3.14.6, NumPy 2.5.4 and PyTorch 2.15.0+cpu. Recorded Torch/BLAS thread controls were one thread. Exact environment details and threadpool metadata are retained in the run manifest. Timings are measurements of these implementations on this machine, rather than hardware-independent complexity estimates. Fitting, CV selection and evaluation costs were recorded separately; three-start neural cost was also summarized. CV selection-plus-selected-fit costs exclude additional nonselected full-fit diagnostics and evaluation.

The archived evidence includes observations and folds, candidate fits and profiles, selected amplitudes, neural checkpoints and traces, complete status records, residual diagnostics, scoring checks and integrity manifests. Dense fields were retained for the first design in each stratum; other fields can be reconstructed from amplitudes or checkpoints. The complete archive is in commit 4a64ed4cae0991bb957c18dd2bcea26162bd4e22 of pinn-audit.

3 Results

3.1 Completion and numerical verification

All 400 scheduled pipeline jobs completed: 160 reported classical pipeline outputs and 240 neural fits across 40 dataset cases. Those cases originated from 20 independent observation/noise designs; family A/B pairs shared a design. Every observation-count/family stratum contained ten designs and every neural case retained all three starts. There were no failed, timed-out or interrupted jobs and no recorded bound hits. All neural fits reached their configured L-BFGS iteration caps. Completion therefore describes a fixed-budget procedure, not certified convergence or a global optimum.

The archived source, run-output and analysis-output hashes matched the integrity records. The planned 401/801-point checks covered forty procedure/start outputs. Their maximum absolute changes were 0.001868 percentage points for field error and approximately 7.04e-8 percentage points for initial-field error. The primary 401-point scores were retained.

A supplementary post-evaluation audit recomputed 2000 held-out CV predictions and their selection decisions, reloaded all 240 neural checkpoints, and verified the 24 saved dense neural fields without retraining. Analytic sine-orthogonality and time-integration checks of all 160 reported classical states agreed with initial-field scores to floating-point precision. The largest field-score discrepancy was 0.1632 percentage points for an unregularized fit whose field error was about 645.5%; the CV12 maximum discrepancy was 0.00321 percentage points. These supplementary checks do not replace the primary analysis.

3.2 Recovery across observation counts and initial fields

Table 1 reports all procedures. Each value is a median percentage error over ten designs. A neural design score first takes the median of its three initialization errors. These are marginal medians; paired effects are reported separately.

Table 1: Marginal median relative errors (%).

N	family	procedure	diffusivity	field	initial field
10	A	unreg6	9.259	10.242	37.274
10	A	fixed6	4.030	3.439	8.508
10	A	CV6	4.030	3.439	8.508
10	A	CV12	4.755	3.942	10.900
10	A	PINN-200	3.919	3.717	8.803
10	A	PINN-1000	2.580	3.842	8.963
10	B	unreg6	9.268	12.976	51.270
10	B	fixed6	6.343	6.187	24.206
10	B	CV6	7.550	6.278	24.206
10	B	CV12	11.752	7.402	26.111
10	B	PINN-200	9.205	6.040	23.810
10	B	PINN-1000	3.400	6.071	23.368
50	A	unreg6	1.325	0.862	2.473
50	A	fixed6	1.326	0.834	2.064
50	A	CV6	1.420	0.834	2.064
50	A	CV12	1.607	0.889	2.919
50	A	PINN-200	1.402	0.760	1.573
50	A	PINN-1000	1.494	0.909	2.366
50	B	unreg6	3.625	5.899	26.250
50	B	fixed6	3.761	5.218	23.362
50	B	CV6	5.526	6.233	24.982
50	B	CV12	1.902	2.191	10.882
50	B	PINN-200	5.356	4.933	22.352
50	B	PINN-1000	2.334	3.703	16.482

Family A contains modes 1 and 3; family B additionally contains mode 7. Unregularized sparse fitting showed substantial initial-field error, while regularization reduced that error in these simulations. At N=10/A, median initial-field error was 37.27% for unreg6, 8.51% for fixed6, 8.51% for CV6, and 10.90% for CV12. The corresponding PINN-200 and PINN-1000 errors were 8.80% and 8.96%. These descriptive comparisons show why the choice of classical regularization matters to an apparent neural advantage.

The denser family-B setting illustrates the importance of basis capacity. CV6 had median initial-field error 24.98%, compared with 10.88% for CV12, 22.35% for PINN-200 and 16.48% for PINN-1000. The richer classical reference therefore changes the comparison suggested by the restricted six-mode baseline. Increasing basis capacity was not uniformly beneficial in the sparse case: at N=10/B, CV12 had median initial-field error 26.11%, compared with 24.21% for CV6. Representation alone did not ensure accurate recovery from limited noisy observations.

3.3 Prespecified paired contrasts against CV12

Table 2 contains all 24 planned contrasts. Differences are neural minus CV12 in percentage points, so negative values favor the neural procedure. Each contrast contains ten complete design pairs. Intervals are pointwise 95% percentile bootstrap intervals for the median paired difference, not simultaneous intervals across outcomes or procedures. Wins are strict error comparisons and are not hypothesis tests.

Table 2: Paired differences against CV12 (percentage points). Wins/ties/losses are for the neural procedure. W/T/L denotes wins/ties/losses.

N	Family	Procedure	Outcome	Median diff.	95% interval	W/T/L
10	A	PINN-200	diffusivity	+0.078	[-12.210, 2.788]	5/0/5
10	A	PINN-200	field	+0.072	[-3.303, 1.391]	4/0/6
10	A	PINN-200	initial field	+0.420	[-8.141, 3.834]	4/0/6
10	A	PINN-1000	diffusivity	-0.135	[-11.756, 0.895]	6/0/4
10	A	PINN-1000	field	-0.382	[-2.983, 0.661]	6/0/4
10	A	PINN-1000	initial field	-0.766	[-7.597, 2.538]	6/0/4
10	B	PINN-200	diffusivity	-1.333	[-14.067, 3.610]	6/0/4
10	B	PINN-200	field	-0.527	[-2.774, 1.585]	6/0/4
10	B	PINN-200	initial field	-0.743	[-4.046, 2.201]	6/0/4

N	Family	Procedure	Outcome	Median diff.	95% interval	W/T/L
10	B	PINN-1000	diffusivity	-4.445	[-22.168, -1.109]	9/0/1
10	B	PINN-1000	field	-1.117	[-3.099, -0.136]	8/0/2
10	B	PINN-1000	initial field	-1.508	[-5.148, -0.467]	9/0/1
50	A	PINN-200	diffusivity	+0.059	[-0.597, 1.016]	4/0/6
50	A	PINN-200	field	-0.058	[-0.261, 0.019]	6/0/4
50	A	PINN-200	initial field	-0.694	[-1.543, -0.216]	9/0/1
50	A	PINN-1000	diffusivity	+0.042	[-0.643, 0.878]	5/0/5
50	A	PINN-1000	field	+0.055	[-0.049, 0.269]	3/0/7
50	A	PINN-1000	initial field	-0.024	[-0.516, 0.884]	5/0/5
50	B	PINN-200	diffusivity	+2.031	[1.209, 4.901]	1/0/9
50	B	PINN-200	field	+2.862	[2.021, 3.562]	0/0/10
50	B	PINN-200	initial field	+11.866	[7.869, 15.369]	0/0/10
50	B	PINN-1000	diffusivity	+0.410	[-0.386, 1.616]	4/0/6
50	B	PINN-1000	field	+0.803	[0.032, 2.142]	2/0/8
50	B	PINN-1000	initial field	+2.577	[-0.303, 11.081]	3/0/7

The strongest favorable co-primary pattern for PINN-1000 occurred at N=10/B. Its diffusivity error was lower than CV12 in nine of ten designs, with median paired difference -4.445 percentage points (interval [-22.168,-1.109]). Initial-field error was lower in nine designs, with median difference -1.508 percentage points (interval [-5.148,-0.467]). The mandatory field outcome also favored PINN-1000 in eight designs, with median difference -1.117 percentage points (interval [-3.099,-0.136]). The wide diffusivity interval and ten-design sample limit the precision of the estimated advantage.

At N=50/B, CV12 had lower field error than PINN-200 in every design and lower diffusivity error in nine designs. For PINN-200 minus CV12, median differences were +2.862 percentage points for field error, +2.031 for diffusivity and +11.866 for initial-field error; all three pointwise intervals were positive. Against PINN-1000, CV12 had lower field error in eight designs, with median paired difference +0.803 percentage points (interval [0.032,2.142]). The longer budget's diffusivity and initial-field intervals spanned zero, so those outcomes did not resolve a direction as clearly as the field comparison.

For family A, all four diffusivity contrasts had intervals spanning zero. At N=50/A, PINN-200 had lower initial-field error in nine designs, with median paired difference -0.694 percentage points (interval [-1.543,-0.216]); other family-A field/initial-field intervals spanned zero. The study therefore supports outcome- and regime-specific comparisons rather than an overall method ranking. Intervals spanning zero do not demonstrate equal performance, and intervals excluding zero should be interpreted in the context of small strata and multiple pointwise comparisons.

3.4 Representation floor, optimizer budget and tail outcomes

For family B, orthogonality of the sine modes implies that every field restricted to modes 1–6 has initial-field relative L2 error at least

$$\frac{0.25}{\sqrt{1^2 + 0.5^2 + 0.25^2}} = \sqrt{1/21} \simeq 21.82\%. \tag{10}$$

All sixty family-B six-mode outputs respected this floor. It describes a known function-space restriction, not a new theorem or evidence that the seventh mode is identifiable from the available observations. The twelve-mode reference contains the truth and hence benefits from a favorable representation assumption.

More L-BFGS iterations did not uniformly reduce all physical recovery errors. For example, at N=50/A the marginal median initial-field error increased from 1.57% for PINN-200 to 2.37% for PINN-1000, while field error increased from 0.76% to 0.91%. At N=50/B, the corresponding medians decreased from 22.35% to 16.48% for the initial field and from 4.93% to 3.70% for the field. These are descriptive budget patterns, not a newly introduced paired budget test, proof of overfitting or a basis for choosing a budget retrospectively using truth.

Median summaries did not capture all large errors. Across individual raw fits, unreg6 reached approximately 645.5% field error and 2622.4% initial-field error; PINN-200 reached 107.8% diffusivity error. All outcomes were retained. The lack of numerical failures does not imply uniformly reliable physical reconstruction or zero failure risk beyond this sample.

3.5 Computational cost

Collection took about 2.66 hours. Table 3 reports median per-design costs for CV12 selection plus the selected fit, and the summed training costs of all three neural starts. These costs correspond to different procedures and are not

an equal-compute comparison. Neural performance summaries use three starts; a single-start time should not be presented as their full computation cost.

Table 3: Median per-design computation costs (seconds), excluding evaluation.

N	Family	CV12 selection + fit	PINN-200 (3 starts)	PINN-1000 (3 starts)
10	A	7.25	73.91	115.24
10	B	9.56	83.70	121.64
50	A	11.05	88.67	124.17
50	B	10.40	95.01	121.62

Single-start neural training medians ranged from approximately 24–41 seconds across strata and budgets. CV12 costs in Table 3 exclude nonselected full-data candidate diagnostics and evaluation; separate timing components are retained in the archive. These machine- and implementation-specific measurements do not establish general timing superiority across hardware or tuned solver families.

4 Discussion

4.1 Interpretation of the comparison

The completed study does not yield a single method ranking across observation counts, initial fields and outcomes. The longer-budget PINN's clearest favorable co-primary pattern against CV12 occurs for the sparse family-B case: diffusivity and initial-field errors are lower in nine of ten designs. In the denser family-B case, CV12 has lower full-field error than the longer-budget PINN in eight of ten designs. Several other contrasts remain uncertain. These results support a conditional comparison of the declared procedures.

This interpretation is narrower than a claim that neural inversion succeeds where numerical inversion cannot. Classical joint recovery already has precedents under different measurement conditions. [1] Existing noisy inverse PINN–numerical comparisons also make the broader benchmarking question established territory. [4], Jekic et al., v2 (2026) Our contribution, if judged sufficient, is the documented sensitivity of the comparison to explicit choices within this small joint heat-inversion problem.

4.2 Representation and regularization

The family-B representation control is essential to interpreting any apparent neural advantage over six-mode inversion. That reference cannot express the seventh sine component, and its initial-field relative error has a 21.82% floor even with ideal coefficients for the retained modes. CV12 removes this particular restriction. Its stronger denser-family-B recovery changes the impression given by CV6 alone. An inadequately expressive comparator can therefore support a different conclusion from a richer one within this benchmark.

The richer representation does not guarantee stronger sparse-data recovery. The contrast between N=10 and N=50 is consistent with a tradeoff between allowed complexity and the information available to constrain it, but our experiment does not uniquely identify that mechanism. Both representation width and the regularization selected from noisy observations affect the resulting fit. Cross-validation targets held-out observation prediction, whereas our physical outcomes include an unobserved initial state and a scalar coefficient.

Heat smoothing provides relevant physical context. In our analytical solution, mode j is attenuated by exp[-alpha(*j*pi)^2*t]; higher-frequency initial detail therefore decays faster. This explains why accurate later temperatures alone need not ensure accurate initial-field reconstruction. It does not establish structural non-identifiability for the chosen data or explain every numerical failure. No uniqueness theorem or complete information analysis was established by these comparisons.

4.3 Optimization budget and the recovery target

Longer training did not uniformly improve physical recovery. The family-A N=50 median initial-field error increased from 1.57% to 2.37% between the two budgets, while the corresponding family-B median decreased from 22.35% to 16.48%. These are descriptive patterns, not a separately tested budget effect or proof of overfitting. Both budgets reached their iteration caps.

The neural loss combines observation discrepancy and PDE residual; it is not the hidden-truth recovery error. A completed optimization run, small residual or favorable single outcome should consequently not replace direct

reporting of diffusivity, initial-state and full-field scores. Retaining all starts also reveals large errors that a median alone would conceal. Our three-start median summarizes an outcome distribution; it is not a deployable ensemble or a model selected without access to truth.

4.4 Limits of inference and fairness

There are ten independent designs in each stratum. The reported bootstrap intervals are pointwise and unadjusted for the 24 contrasts. They should not be read as simultaneous evidence of method-class superiority, and intervals spanning zero do not establish equivalence. Family A/B cases share randomness; the study contains twenty independent designs rather than forty independent replications. Neural initializations further reuse observations and do not increase the number of independent data designs.

The twelve-mode representation contains the generating truth, and the classical time dependence satisfies the specified PDE exactly. This is favorable knowledge relative to an unrestricted neural field. Conversely, classical regularization was selected by cross-validation while neural architecture and weights were fixed from development. These differences prevent a claim of matched priors, equally tuned method classes or equal computation. Timing comparisons also require distinct accounting for CV selection, selected fitting, additional diagnostics, evaluation and all three neural starts.

Only one diffusivity, one noise level, one spatial domain and two finite spectral initial states were investigated. Noise was independent additive Gaussian error, PDE and boundary assumptions were correct, and the data were synthetic. Fixed collocation and three initialization seeds do not capture all neural training variability. Application to real measurements, model mismatch, other noise distributions or more complex domains remains untested. Numerical verification supports the recorded scores, not generalization beyond this setup.

4.5 What follows from this study

A defensible report should present the full result matrix and state exactly which conclusions change when comparator representation, regularization or neural budget changes. Its value would lie in a transparent, inspectable example of comparative sensitivity, rather than a new architecture or a broad performance claim. Whether that example adds enough to the existing literature remains an editorial and scientific judgment.

Future investigation could examine non-spectral initial fields, multiple noise levels and diffusivities, model mismatch, or explicitly balanced tuning budgets. Those questions require a new protocol and fresh evaluation data. Retuning the current cases would make them development evidence for a new procedure; their present outcomes should remain unchanged and fully archived.

5 Conclusion

This controlled study compared specified neural and spectral procedures for joint diffusivity and unknown-initial-field heat inversion. Explicit regularization and representation controls changed the context of apparent neural advantages, while retaining both neural budgets revealed that longer optimization did not uniformly improve physical recovery. The strongest favorable co-primary pattern for the longer neural budget occurred with sparse observations of the richer initial field; denser observations favored CV12 for the full-field comparison. Other contrasts remained uncertain.

These findings concern one synthetic heat problem, two finite spectral initial states and fixed procedural choices. They establish neither a new solver nor general superiority of a method class. The versioned protocol and complete artifact archive support inspection of the conditional findings, including unfavorable and uncertain outcomes. Broader conclusions require fresh evaluation under additional physical, data and tuning conditions.

Statements and declarations

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Competing interests

The author declares no competing interests.

Data and code availability

The source code, synthetic observations, configurations, fitted states, training traces, analysis and integrity records are available in the pinn-audit repository. The completed experiment archive is preserved at commit `4a64ed4cae0991bb957c18dd2bcea26162bd4e22`. Later documentation and CI fixes do not change that archived experiment.

Use of AI assistance

Claude and ChatGPT assisted with developing and debugging Python code for experiments and statistical analysis. Numerical calculations were performed by Python scripts. The author reviewed the references, prepared and revised the manuscript, and retains responsibility for the accuracy and integrity of the final work.

Author contributions and approval

The contribution statement and final author approval remain to be completed after the author list is settled and each author has reviewed the manuscript. No assertion of completed human verification or approval is made here.

Research materials

This study uses synthetic observations of a mathematical heat model. No human participant data, animal experiments or clinical measurements were collected for this study. The MRI studies discussed in related work are prior literature, not datasets analyzed in the present experiment.

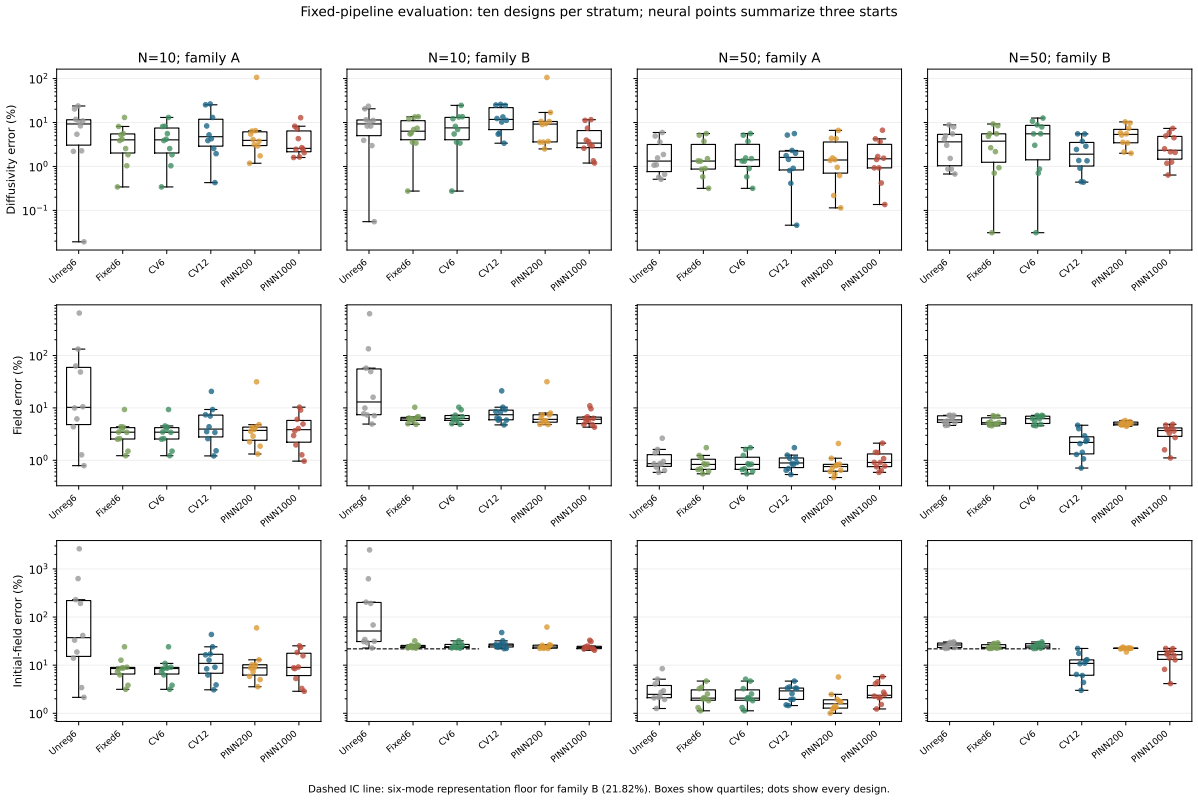


Figure 1: Errors across all six procedures. Each point represents one design-level error. Neural points summarize the three starts separately for each outcome; they are not ensemble-model predictions. Errors use a logarithmic vertical scale. The six-mode family-B initial-field floor is a representation limit. Read together with Table 1, including tails.

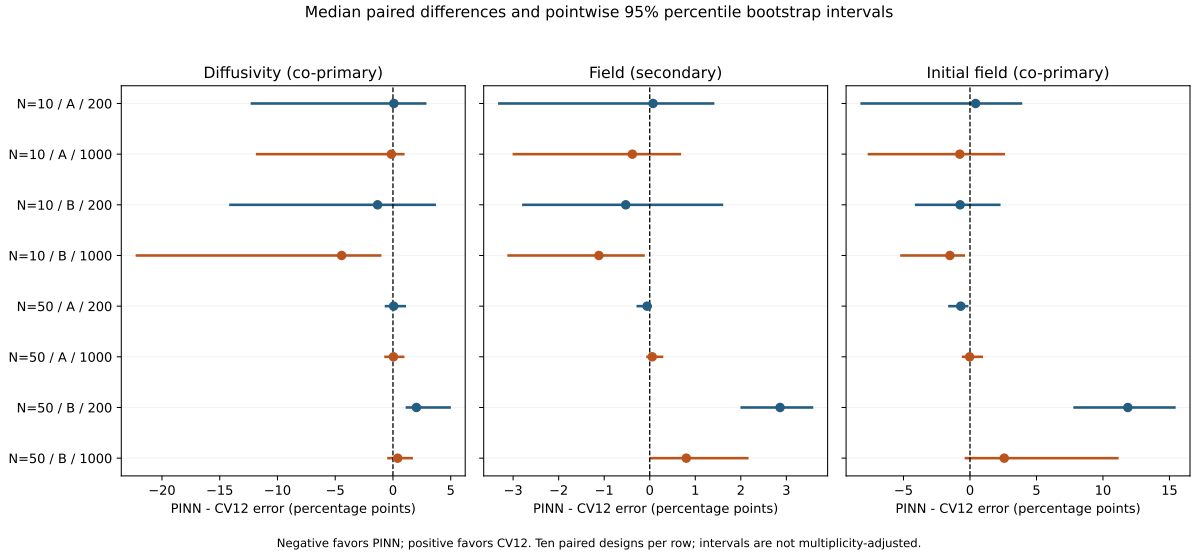


Figure 2: All planned paired contrasts against CV12. Differences are PINN minus CV12 in percentage points: negative favors PINN. Intervals are pointwise 95% percentile bootstrap intervals for median paired differences across ten designs per condition, with no multiplicity adjustment. They are not error bars on the individual observations or simultaneous intervals.

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